

Topic 10

RRH (2017): Counterfactuals

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Quantitative Spatial Economics

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Summer term 2025

Acknowledgements

- ▶ This slide deck uses material from the following lectures
 - ▶ Quantitative Spatial Economics lecture in Princeton University course EC552—by Esteban Rossi-Hansberg

Course evaluation

Quantitative Spatial Economics (2025)

CourseSettingsParticipantsGradesReportsMore ▾

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General

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Forum

Syllabus +++ includes instructions on assessment +++ PDF

Assessed work, submission guidelines, MSc students PDF

Assessed work, submission guidelines, PhD students (BSGE) PDF

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Topic 10 RRH (2017): Counterfactuals

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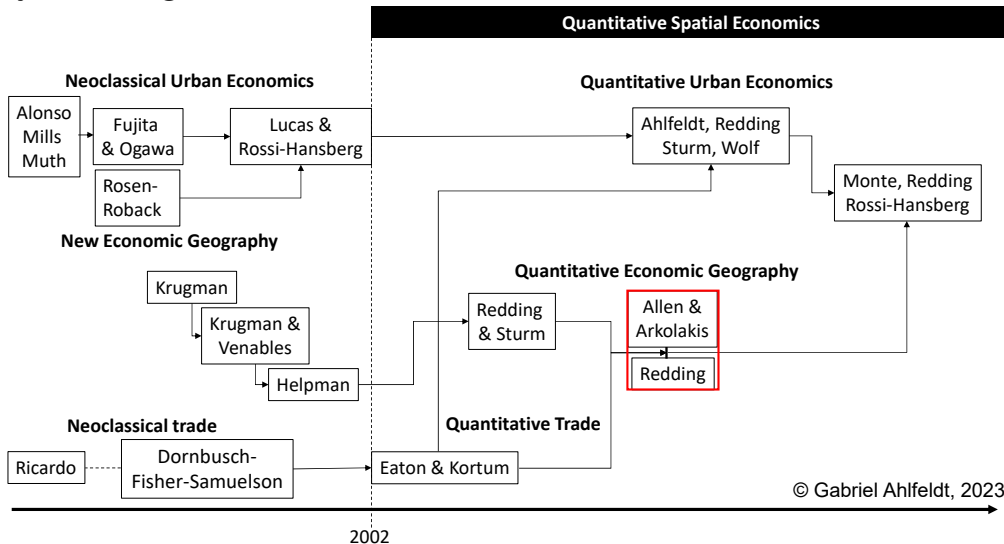
Introduction

Quantitative Spatial Models of Economic Geography

- ▶ Quantitative spatial models of economic geography emphasize **trade of goods**
 - ▶ Low trade costs to other locations $\Rightarrow$ Greater market access
 - ▶ Greater market access $\Rightarrow$ lower price index (and higher import/export shares)
 - ▶ Lower tradable prices $\Rightarrow$ attract larger population
 - ▶ Larger population $\Rightarrow$ higher non-tradable goods prices in **spatial equilibrium**
 - ▶ Inelastic supply of land generates dispersion force
 - ▶ Higher non-tradable prices compensate for lower tradable goods prices

Q: How will the economic geography change if we reduce trade frictions?

History of thought



Roadmap

- ▶ **Redding-Rosihansberg (2017) model**
 - ▶ A canonical QSM of Economic Geography
 - ▶ Multi-region version of the Helpman (1998) model
 - ▶ Also used in Redding & Sturm (2008)
- ▶ **Topic 9**
 - ▶ Model
 - ▶ Equilibrium
 - ▶ Quantification
- ▶ **Topic 10 (today)**
 - ▶ Counterfactuals

Equilibrium solver

Recall

- ▶ The equilibrium condition uses
 - ▶ the following equations: Trade share (9), price index (8), population mobility (13)
 - ▶ the assumption that trade costs are symmetric ($d_{ni} = d_{in}$)
- ▶ The equilibrium is referenced solely by L_n
 - ▶ Can solve for L_n from Eq. (16) for given values of exogenous objects
 - ▶ structural parameters $\{\alpha, \sigma\}$, fundamentals $\{A_n, H_n\}$, trade costs d_{ni} , fixed cost, F
 - ▶ We do not need to solve for any other endogenous variable simultaneously
 - ▶ there is a **recursive structure** to solve for the other endogenous objects
- ▶ Can treat L_n as our sole target variable and then solve for the rest

Sounds good in theory, in practice...

In practice...

- ▶ ...RRH treat $\{L_n, w_n\}$ as target variables
 - ▶ Guess them
 - ▶ Solve simultaneously
- ▶ Lesson for us:
 - ▶ Nice to show that the equilibrium can be referenced by one variable
 - ▶ If the solver is easier to write with two target variables, we may still do it...

```
Editor - /usr/net/ahlfeldg/Teaching/RRH2017-ARE/functions/solveHLwCtyOpen_E.m
calculateHHI.m GA_QSE_RRH2017_teaching.m solveHLwCtyOpen_E.m +
1  %% Solve model
2
3  function [w_i, L_i, tradesh, dtradesh, converge, xtic] = solveHLwCtyOpen_E(fund, dist, bord, bordc, nobs)
4
5  global alpha sigma LL LLwest LLeast;
6
7  xtic = tic();
8
9  % Extract location characteristics from fundamentals matrix;
10 % fund(:,1)=a; fund(:,2)=H; fund(:,3)=Iwest; fund(:,4)=Ieast;
11 a=fund(:,1); H=fund(:,2); Iwest=fund(:,3); Ieast=fund(:,4);
12
13 % convergence indicator;
14 converge=0;
15
16 % Initialization based on a symmetric allocation;
17 L_i=double(ones(nobs,1)).*(LL./nobs);
18 w_i=double(ones(nobs,1));
19
20 % trade costs;
21 dd=double((dist.*bord.*bordc).^(1-sigma));
22
23 % *****;
24 % **** START LOOP TO SOLVE FOR WAGES AND POPULATION ****;
25 % *****;
```

What seems noteworthy...

- ▶ Since RRH are targeting $\{L_n, w_n\}$, they need updating rules for both
 - ▶ L_n is relatively straightforward
 - ▶ predict own trade shares π_{nn} for guesses of L_i (and w_i) using Eq. (9)
 - ▶ get $\lambda \bar{L}$ for π_{nn} using population share Eq. (15)
 - ▶ Can use that to update guess of L_n until we converge
 - ▶ As for w_n , RRH mention:
 - ▶ Zero profits $\Rightarrow$ wage bill equals firm revenue
 - ▶ Rent redistribution only rescales worker income by $1/\alpha$
 - ▶ *Income equals expenditure:*

$$\underbrace{w_i L_i}_{\substack{\text{worker income} \\ = \text{firm wage bill}}} = \underbrace{\sum_{n \in N} \pi_{ni} w_n L_n}_{\substack{\text{expenditure by all workers} \\ \text{on goods from } i \\ = \text{firm sales revenue}}}$$

But how does this help with updating w_n ?

Income equals expenditure

- ▶ RRH compute expenditure using trade shares in `solveHLwCtyOpen_E.m`
 - ▶ First compute `income=double(w_i.*L_i)`
 - ▶ Worker **income** (equivalently, the firm's wage bill)
 - ▶ Then compute `expend=double(tradesh*income)`
 - ▶ Total **expenditure** by all locations on goods produced in i
 - ▶ π_{ni} is the share of location n 's expenditure on goods produced in location i (in the MATLAB tradeshare matrix, n are columns, i are rows)
- ▶ If expenditure $>$ ($<$) income, we need to increase (reduce) the wage
 - ▶ `w_e=double(w_i.*(expend./income).^(1./(sigma-1)))`
 - ▶ w up $\Rightarrow$ income up, expenditure down (share in consumption of other regions will fall)
 - ▶ Higher wages $\Rightarrow$ higher prices $\Rightarrow$ lower demand
 - ▶ The exponent $1/(\sigma - 1)$ translates the desired reduction in expenditure into the required wage (price) increase under CES demand

Exact hat algebra

Conducting counterfactuals

- ▶ The standard approach to counterfactuals (for given parameters) in QSMs is
 - ▶ **Invert** unobserved **exogenous objects** using observed data
 - ▶ e.g. location fundamentals, transport costs
 - ▶ **Change** values of selected **primitives**
 - ▶ e.g. productivity in some regions, transport cost on some routes
 - ▶ **Solve** for the equilibrium
 - ▶ Under the new primitives
 - ▶ Compare new equilibrium values to initial values and **compute relative changes**

Can we skip inversion? We are using observed data for inversion anyway...

Exact hat algebra

- ▶ Dekle et al. (2007) denote the
 - ▶ known value of a variable in the initial equilibrium by x
 - ▶ unknown value of a variable in the counterfactual equilibrium by x' (with a prime)
 - ▶ relative change in the variable by $\hat{x} = \frac{x'}{x}$ (with a hat)
- ▶ We can compute the counterfactual value $x' = \hat{x}x$
 - ▶ We only need the relative changes and initial levels of a variable
- ▶ Aim of **exact hat algebra** is
 - ▶ express **relative changes in endogenous variables** as functions of
 - ▶ **relative changes in primitives**
 - ▶ **initial values of endogenous variables**

We can derive system of equations that avoid levels of fundamentals altogether

Example

- ▶ RRH assume that researcher
 - ▶ observes $\{L_n, w_n, \pi_{ni}\}$
 - ▶ has obtained estimates of the values of structural parameters α, σ
 - ▶ have computed a measure of relative change in bilateral trade costs $\hat{d}_{ni}$
- ▶ They derive a system of equations to evaluate the effects of changes in trade cost
 - ▶ combining Eqs. (9), (10), (11), (12), (13)
- ▶ System of Eqs. (18), (19), (20) has **three equations and three unknowns** ✓
 - ▶ Can use it to solve for $\{\hat{w}_n, \hat{\lambda}_n, \hat{\pi}_{ni}\}$ using only $\{\lambda_n, w_n, \pi_{ni}, \hat{d}_{ni}\}$
 - ▶ Counterfactual values in levels are $w'_n = w_n \hat{w}_n$, $\lambda'_n = \lambda_n \hat{\lambda}_n$, $\pi'_{ni} = \pi_{ni} \hat{\pi}_{ni}$
 - ▶ Recall that $L_n = \lambda_n \bar{L}$

Exact hat system of equations

- ▶ $\hat{w}_i \hat{\lambda}_i (w_i \lambda_i) = \sum_{n \in N} \hat{\pi}_{ni} \hat{w}_n \hat{\lambda}_n \pi_{ni} (w_n \lambda_n)$
 - ▶ 'hat' variant of **income equals expenditure** $w_i L_i = \sum_n \pi_{ni} w_n L_n$
 - ▶ where $L_n = \lambda_n \bar{L}$ and $\bar{L}$ drops out in ratios
- ▶ $\hat{\pi}_{ni} \pi_{ni} = \frac{(\hat{d}_{ni} \hat{w}_i)^{1-\sigma} \hat{L}_i \pi_{ni}}{\sum_{k \in N} (\hat{d}_{nk} \hat{w}_k)^{1-\sigma} \hat{L}_k \pi_{nk}}$
 - ▶ 'hat' variant of **trade shares**
 - ▶ Notice that $\hat{L}_i$ could be written as $\hat{\lambda}_i$ and A_i drops out in ratios
- ▶ $\hat{\lambda}_n \lambda_n = \frac{\hat{\pi}_{nn}^{-\frac{\alpha}{\sigma(1-\alpha)-1}} \lambda_n}{\sum_{k \in N} \hat{\pi}_{kk}^{-\frac{\alpha}{\sigma(1-\alpha)-1}} \lambda_k}$
 - ▶ 'hat' variant of of residential choice probability equation
 - ▶ Notice that $\{A_n, H_n\}$ drop out in ratios

Welfare

- ▶ Welfare is given by the population mobility condition: $V_n = \frac{v_n}{P_n^\alpha r_n^{1-\alpha}} = \bar{V}$ (Eq. 13)
 - ▶ $v_n = w_n / \alpha$ (Eq. 11)
 - ▶ $P_n = \frac{\sigma}{\sigma-1} \left(\frac{L_n}{\sigma F \pi_{nn}} \right)^{\frac{1}{1-\sigma}} \frac{w_n}{A_n}$ (Eq. 10)
 - ▶ $r_n = \frac{1-\alpha}{\alpha} \frac{w_n L_n}{H_n}$ (Eq. 12)
- ▶ Plug into Eq. (13) and observe the magic of 'exact hat algebra'
 - ▶ $\hat{\bar{V}} = \frac{\bar{V}'}{\bar{V}} = \left(\frac{1}{\hat{\pi}_{nn}} \right)^{\frac{\alpha}{\sigma-1}} \left(\frac{1}{\hat{\lambda}_n} \right)^{\frac{\sigma(1-\alpha)-1}{\sigma-1}}$ (Eq. 21)
 - ▶ Various primitives A_n, H_n, F and multiplicative constants cancel out in ratios

Merits of exact hat algebra

- ▶ Exact hat algebra allows expressing counterfactual values with **fewer primitives**
 - ▶ **Simplifies notations**
 - ▶ Can drop exogenous objects $\Rightarrow$ leaner equations
 - ▶ **Counterfactuals more transparent**
 - ▶ Link between forcing variable and outcome variables in relative differences
 - ▶ Can **save a space** in the paper as we can skip inversion
 - ▶ QSE papers tend to be too long $\Rightarrow$ Issue in publication process
 - ▶ Still worth doing the usual **sanity check**
 - ▶ Can we recover the **initial equilibrium from inverted** fundamentals...

Q: How do RRH use exact hat algebra in their code?

Counterfactuals

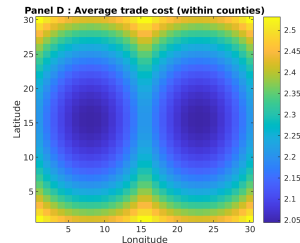
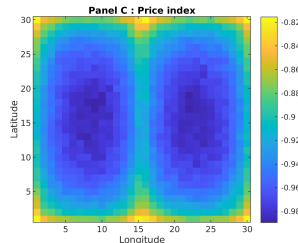
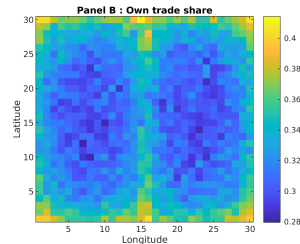
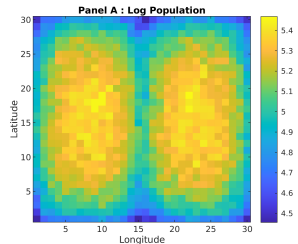
Replication directory

You can replicate the following counterfactuals using the RRH replication directory with GA edits available from [Moodle](#)
Follow the GA_QSE_RRH2017_teaching.m script

Recall

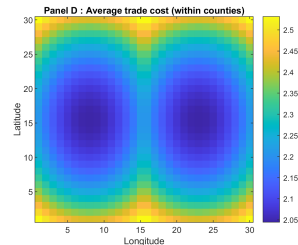
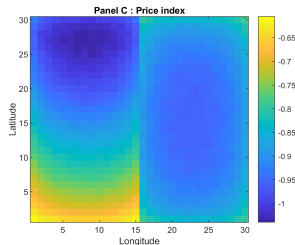
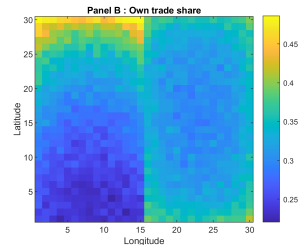
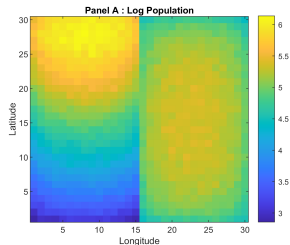
- ▶ RRH model with GA's parameterization of A_n
- ▶ Central locations have lower average trade costs (within countries)
- ▶ Results in more trade (lower own share) and a lower price index
- ▶ And, indeed, a greater population

Q: What will happen if we increase productivity in the north of the western country?



With productivity gradient

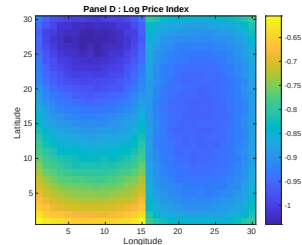
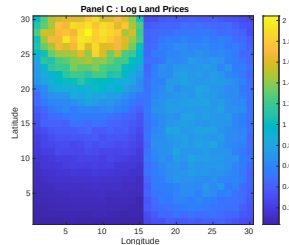
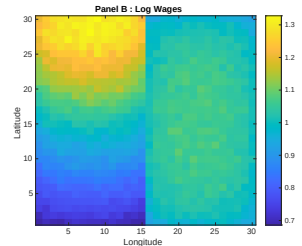
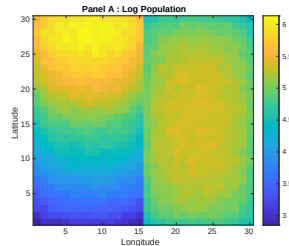
- ▶ Higher productivity in the north of west, shifts labour demand outwards
- ▶ Larger population in the north of west
- ▶ Own trade share increases in the north of west
 - ▶ not because places are more remote
 - ▶ they are larger and sell more to themselves $\Rightarrow$ 'home market effect'



Initial equilibrium

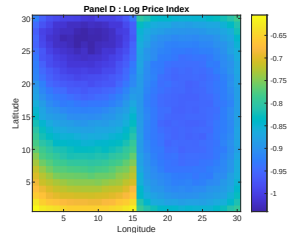
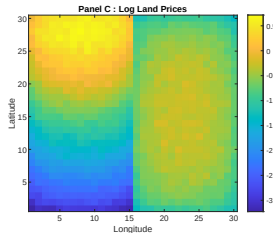
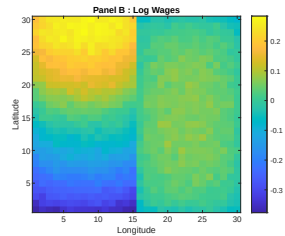
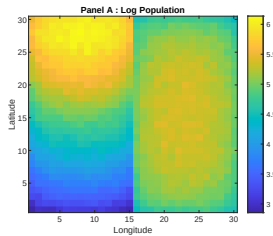
- ▶ Higher productivity in the north of west:
 - ▶ higher wages
 - ▶ higher rents
- ▶ West has **more unequal** distribution of productivity
 - ▶ Also a **more unequal** distribution of pop.
 - ▶ HHI West: 34.7%
 - ▶ HHI East: 23%

Q: What happens when removing border frictions **wit-
hin** countries?

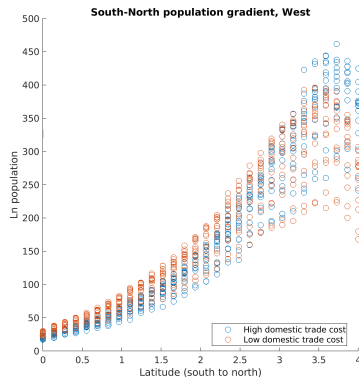


No domestic borders

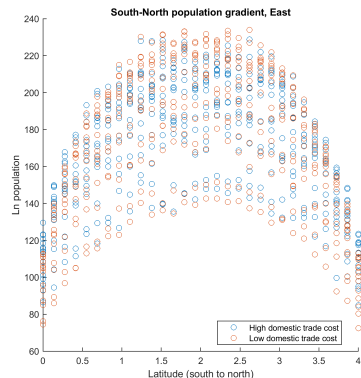
- ▶ Iceberg border friction between grid points now 1 instead of 2
 - ▶ Within East and West
 - ▶ Friction between East and West persists
- ▶ Agglomeration force weaker
⇒ less inequality in West
 - ▶ HHI West: 31.1% (-3.6pp)
- ▶ Hardly any change in East
 - ▶ HHI East: 23.4% (+0.4pp)



Gradients with and without domestic borders



Agglomerated north loses pop. in **west**
Weaker agglo force $\Rightarrow$ less inequality



No big changes in the **east**
Perhaps a bit more dispersion

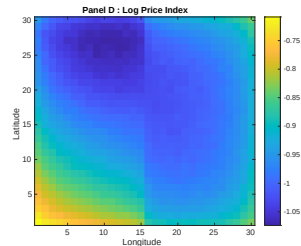
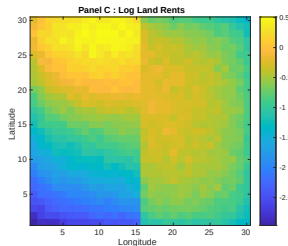
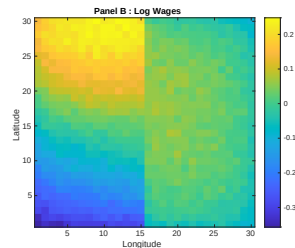
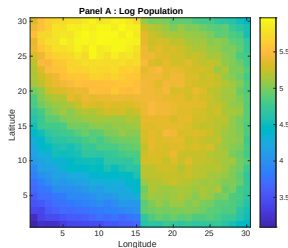
Trade costs and agglomeration

- ▶ **High trade costs reinforce fundamental productivity advantages**
 - ▶ Market access generates an agglomeration force
 - ▶ Consumers benefit from access to products
 - ▶ Firms benefit from access to consumers
 - ▶ With higher trade costs, stronger agglomeration effect

Q: What happens when we remove border frictions **between** countries?

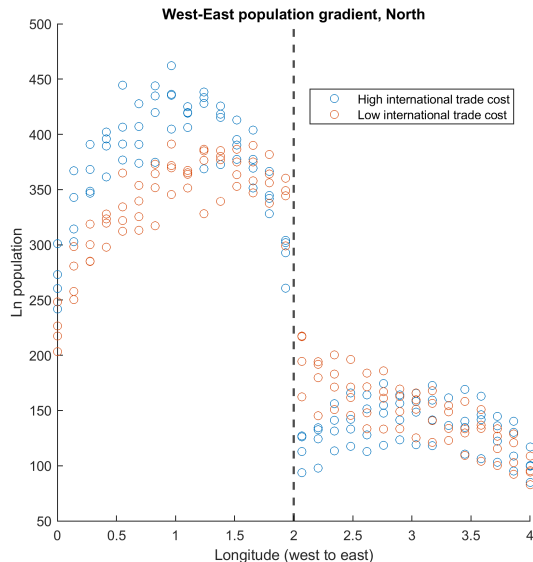
No international borders

- ▶ **Economic activity shifts towards the international border**
 - ▶ Formerly remote areas now have the greatest market access
- ▶ Positive welfare effect: 3.4%
 - ▶ All regions benefit from **greater market access**
 - ▶ Great market access $\Rightarrow$ smaller own trade share
 $\pi_{nn} \Rightarrow$ greater $\bar{V}$ (see Eq. 21)



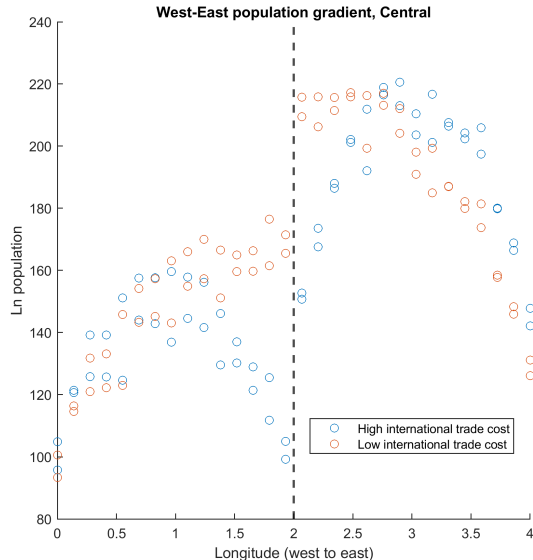
No international borders: North

- ▶ Economic activity shifts towards the international border
- ▶ **'Dip' disappears** on both sides of the border
- ▶ In the less productive east, population density increases close to the border
 - ▶ Locations in the north-east benefit from market access to the productive north-west
- ▶ Within the West, the North loses population since the comparative advantage relative to the South (in the West) shrinks
 - ▶ South gains more market access



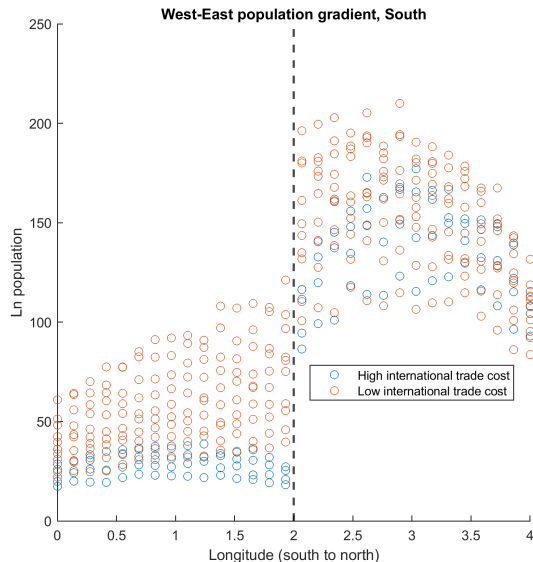
No international borders: Central

- ▶ Productivity is larger in the west, but central parts of east are more populated
 - ▶ In the west, fundamental productivity advantage attracts workers to north
 - ▶ In the east, market access advantage attracts workers to central parts
- ▶ Without border friction, **border dip disappears**



No international borders: South

- ▶ Productivity is similar on both sides of the border
- ▶ But east has a greater market access since much of the population in the west is in the north)
- ▶ Without border friction
 - ▶ **border dip disappears**
 - ▶ Population **density gradient emerges in the west** since areas close to east benefit from market access
- ▶ South in the West benefits since market access mitigates productivity disadvantage relative to North



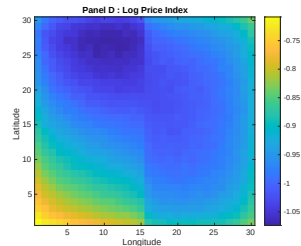
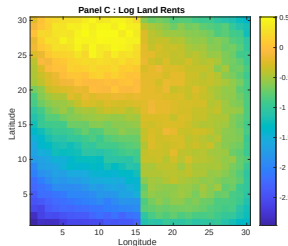
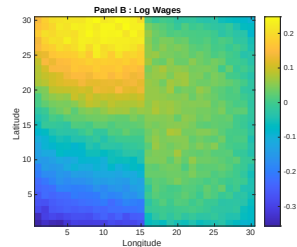
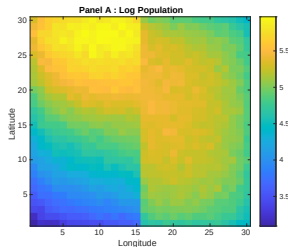
Market access and border effects

- ▶ **Removal of border frictions turns remote areas into central areas**
 - ▶ Increase in market access on both sides of the border $\Rightarrow$ Local economic development
 - ▶ Effect particularly large on the side of the border with initially lower market access
- ▶ **Compelling evidence** supports the role of market access
 - ▶ Evidence from German Division and Reunification: Redding & Sturm (2008)
 - ▶ Evidence from EU integration: Caliendo et al. (2021)

Q: What's the international border effect under lower domestic trade frictions?

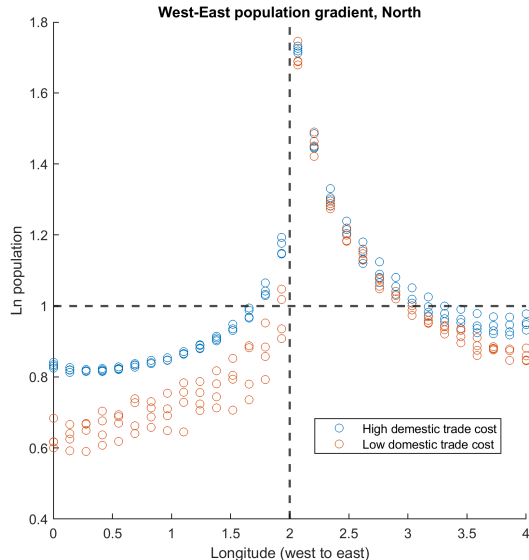
No international borders

- ▶ Same experiment as before, but **without border frictions between grid cells**
 - ▶ No domestic border friction in initial equilibrium and counterfactual
 - ▶ Only international border friction changes
- ▶ Similar spatial adjustments
 - ▶ Economic activity shifts towards the international border
- ▶ **Larger positive welfare effect: 5% (instead of 3.4%)**



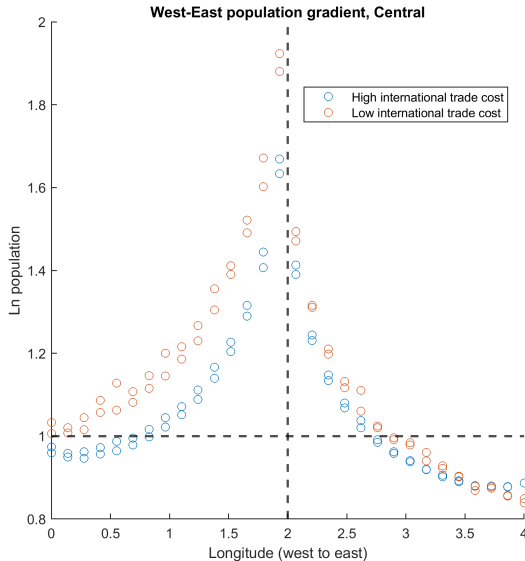
No international borders: North

- ▶ Economic activity shifts towards the international border
- ▶ **Stronger adjustments** under low domestic trade costs
 - ▶ Loss of economic activity in west is GE effect stronger gains in MA in central and southern parts of west



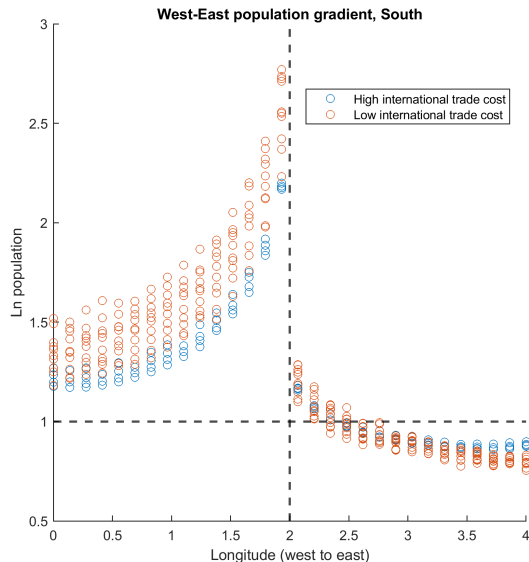
No international borders: Central

- ▶ Economic activity shifts towards the international border
- ▶ **Stronger adjustments** under low domestic trade costs



No international borders: South

- ▶ Economic activity shifts towards the international border
- ▶ **Stronger adjustments** under low domestic trade costs



Conclusion

Summary

- ▶ **QSMs of Economic Geography** emphasize the trade between regions
 - ▶ Market access (MA) plays a crucial role
 - ▶ Reduces tradable goods price index and attracts workers
 - ▶ Prediction that market access promotes economic development has empirical support
- ▶ In reduced-form, **trade MA works similar to commuting MA**
 - ▶ At a small geographic (within cities) commuting MA clearly more important
 - ▶ At a large geographic scale (between labour markets) trade MA more important
 - ▶ At intermediate geographic scale, both could matter...

Next week: **Commuting and trade in one model**

Literature I

Core readings

- ▶ Redding, S., Rossi-Hansberg, E. (2017): Quantitative Spatial Economics. *Annual Review of Economics*, 9, 21–58.

Other readings

- ▶ Caliendo, L., Opromolla L. D., Parro, F., Sforza, A. (2021): Goods and Factor Market Integration: A Quantitative Assessment of the EU Enlargement. *Journal of Political Economy*, 129(12)
- ▶ Dekle R., Eaton J., Kortum S. (2007): Unbalanced trade. *American Economic Review*, 97(2):351–55
- ▶ Redding, S., Sturm, D. (2008): The Costs of Remoteness: Evidence from German Division and Reunification. *American Economic Review*, 98(5), 1766–97